

MarketPulse Funds: Out-of-Sample Multi-Asset Portfolio Design and Finance-Aware News Sentiment

Executive Overview

MarketPulse Funds is a Streamlit research app for systematic equity, cryptocurrency and combined funds. It features 12 baseline funds using equal weight, minimum volatility, tangency and risk parity. It also includes Sentiment Tilt Equity and MarketPulse TailGuard. All funds are tested over 2021–2023 with monthly walk-forward backtests using only information available before each rebalance.

The results reveal distinct trade-offs. Crypto Minimum Volatility achieved the highest return and Sharpe ratio, but suffered a –74.47% maximum drawdown. Among the equity and combined funds, Combined Tangency produced the highest return, while Combined Risk Parity achieved the highest Sharpe ratio. Finance-Extended VADER only marginally improved manual-validation accuracy, while the fixed sentiment tilt reduced return and Sharpe ratio across all four tested methods. TailGuard also failed to outperform Combined Minimum Volatility in realised downside protection. I treat the extensions as useful tests rather than evidence that more complex models necessarily perform better out of sample.

1. Funds and Out-of-Sample Backtest Design

1.1 Fund Menu and Target User

The app is intended for self-directed retail investors and research users looking to compare transparent, rules-based funds and test a hypothetical allocation. It is presented as a research prototype rather than personal financial advice.

The fund menu contains 12 baseline funds across equity, cryptocurrency and combined universes using equal weight, minimum volatility, tangency and risk parity. Two additional funds bring the total to 14. Sentiment Tilt Equity adjusts equity weights using lagged sector news, while MarketPulse TailGuard minimises historical CVaR and caps aggregate cryptocurrency exposure. I regard both extensions as experiments that should be compared with simpler funds rather than assumed to be better in advance.

1.2 Walk-Forward Backtest Design

The funds are evaluated in a monthly walk-forward out-of-sample backtest. On the first available date of each month, portfolio weights are estimated using only returns observed before that date. These are held until the next monthly rebalance. This separation is required so that realised holding-period returns cannot affect the

portfolio decision.

Equity-only and combined funds use a trailing window of 252 observations and are evaluated from 4 January 2021 to 29 December 2023. Crypto-only funds use 365 observations and are evaluated from 1 January 2021 to 31 December 2023. Data from 2020 provide the initial estimation history, while the cryptocurrency observations dated 1 January 2024 are excluded.

Equity and cryptocurrency returns are computed separately on their native calendars. For the combined panel, the completed cryptocurrency daily returns are left-merged onto the equity trading calendar. Weekend-only cryptocurrency movements are thus dropped rather than carried into Monday; the Monday observation contains only the native Sunday-to-Monday return. The crypto-only funds use the complete seven-day calendar. I follow this convention since it is explicitly required by the Project Brief and ensures that each combined fund is evaluated using the same calendar rule.

1.3 Portfolio Construction and Evaluation Rules

All portfolios are long-only and fully invested. At each rebalance, an asset is eligible if it has valid returns for at least 80% of the estimation window. Missing observations for eligible assets are treated as zero, assets with zero estimated volatility are removed, and no individual weight may exceed 20%.

Equal weighting assigns the same weight to every eligible asset:

$$w_{i,t} = \frac{1}{N_t} \quad (1)$$

Minimum-volatility optimisation follows the mean–variance framework of Markowitz (1952) and selects the portfolio with the lowest estimated variance:

$$\min_{\mathbf{w}_t} \mathbf{w}_t^T \Sigma_t \mathbf{w}_t \quad (2)$$

Tangency optimisation selects the portfolio with the highest estimated excess return per unit of estimated volatility, using the Sharpe measure as its objective (Sharpe 1966):

$$\max_{\mathbf{w}_t} \frac{\mathbf{w}_t^T \mu_t - r_f}{\sqrt{\mathbf{w}_t^T \Sigma_t \mathbf{w}_t}} \quad (3)$$

Risk parity seeks to make each eligible asset's contribution to estimated portfolio variance as similar as possible (Maillard, Roncalli & Teïletche 2010):

$$\min_{\mathbf{w}_t} \frac{\sum_{i=1}^{N_t} \left(RC_{i,t} - \frac{\sigma_{p,t}^2}{N_t} \right)^2}{\left(\sigma_{p,t}^2 \right)^2} \quad (4)$$

where $RC_{i,t} = w_{i,t} (\boldsymbol{\Sigma}_t \mathbf{w}_t)_i$ is asset i 's contribution to estimated portfolio variance.

All four methods are subject to the common portfolio constraints:

$$\begin{aligned} \mathbf{1}^T \mathbf{w}_t &= 1, \quad 0 \leq w_{i,t} \\ &\leq 0.20 \end{aligned} \quad (5)$$

In Equations (1)–(5), w denotes the portfolio-weight vector, N is the number of eligible assets, μ is the annualised mean-return vector, Σ is the annualised covariance matrix, and r_f is the annual risk-free rate, set to zero. RC denotes an asset's contribution to portfolio variance. A 10% diagonal shrinkage is applied to Σ to reduce estimation instability in the rolling window.

The resulting out-of-sample portfolio return is calculated using weights estimated strictly before the return observation:

$$\begin{aligned} r_{p,\tau} &= \mathbf{w}_t^T \mathbf{r}_{\tau}, \quad \tau \\ &\in [t, t + 1) \end{aligned} \quad (6)$$

Here, $\mathbf{r}_f$ denotes the vector of asset returns during the holding period after rebalance date t . The final diagnostics contain 612 optimisation records with no solver failure or fallback. TailGuard's CVaR objective and cryptocurrency-sleeve cap are described in Section 4.4.

Performance is compared using growth of one dollar, annualised return and volatility, Sharpe and Sortino ratios, 95% VaR and CVaR, maximum drawdown and total return. The Sortino ratio focuses specifically on downside rather than total volatility (Sortino and van der Meer, 1991). Equity-only and combined funds use 252 periods per year and crypto-only funds use 365. VaR and CVaR are reported as positive daily loss magnitudes. Transaction costs, taxes, slippage and market impact are excluded.

2. Out-of-Sample Results and Fund Fact Sheets

2.1 Performance Comparison

Table A1 compares the 2021–2023 out-of-sample performance of the 14 core funds. Equity-only and combined funds use 252 periods per year, while crypto-only funds use 365. The Sharpe ratio uses a zero risk-free rate.

Crypto Minimum Volatility produced the highest annualised return of 58.40% and the highest Sharpe ratio of 0.986, but its maximum drawdown still reached –74.47%. The practical issue is that a high Sharpe ratio does not eliminate the risk of a large interim loss. The cryptocurrency funds therefore should not be compared against each other by return or Sharpe ratio.

Within the equity and combined funds, Combined Tangency produced the highest annualised return of 19.06%, while Combined Risk Parity produced the highest Sharpe ratio of 0.893. Combined Risk Parity produced a lower return but also lower volatility and shallower drawdown. Equity Equal Weight also remained competitive, producing a higher return and Sharpe ratio than the optimised equity funds. I view this as evidence that a simple benchmark can be helpful even when more complex methods are based on estimates from a relatively short historical window.

Equity Minimum Volatility and Combined Minimum Volatility recorded the shallowest maximum drawdowns, at approximately –15.6%, but their annualised returns were also only about 5.8%. TailGuard produced a lower return and Sharpe ratio than Combined Minimum Volatility. This shows that reducing estimated volatility or tail risk can improve some dimensions of risk but does not guarantee the strongest result overall. TailGuard is discussed in more detail in Section 4.4.

2.2 Growth of One Dollar and Drawdown Analysis

Figure A1 shows the value of one dollar invested in each of the 14 core funds during the out-of-sample period. The funds are divided by asset universe so that the much larger cryptocurrency values do not obscure differences among equity and combined funds. The growth paths show when gains and losses occurred rather than only the result.

The cryptocurrency funds achieved the highest ending values but followed unstable paths. Crypto Minimum Volatility finished at approximately \$3.97 for each initial dollar, while the other crypto funds finished between approximately \$2.03 and \$2.63. Large gains during 2021 were followed by substantial declines, so the ending values alone hide much of the holding-period risk.

Equity Equal Weight finished at approximately \$1.43. Combined Tangency reached approximately \$1.68, compared with about \$1.48 for Combined Risk Parity and \$1.19 for Combined Minimum Volatility. TailGuard finished at approximately \$1.10. The

lower-volatility funds generally followed steadier paths but participated less in the strongest return periods, so I do not treat ending wealth as a complete measure of performance.

Figure A2 shows each fund's decline from its previous running peak, capturing losses that cannot be seen from final wealth alone.

The cryptocurrency funds experienced the deepest drawdowns. Crypto Equal Weight fell by -81.60%, followed by Crypto Risk Parity at -79.94%, Crypto Tangency at -77.12% and Crypto Minimum Volatility at -74.47%. Even the cryptocurrency fund with the highest Sharpe ratio therefore required an investor to tolerate a loss of almost three quarters from a previous peak.

Equity Minimum Volatility and Combined Minimum Volatility recorded the shallowest maximum drawdowns, at -15.62% and -15.77%. Combined Risk Parity reached -19.98%, compared with -23.14% for Combined Tangency. TailGuard's maximum drawdown was -18.37%, providing some protection relative to Tangency and Equal Weight but not outperforming Combined Minimum Volatility.

2.3 Portfolio Weights and Diversification

Figure A3 compares the total cryptocurrency weight of the four baseline combined funds and TailGuard at each monthly rebalance. All five funds use the same 60-asset universe, but their optimisation methods produce very different exposures to cryptocurrency.

Combined Equal Weight maintained a constant cryptocurrency allocation of 16.67%. Combined Minimum Volatility allocated an average of only 0.72% to cryptocurrency, while Combined Risk Parity maintained a more stable average allocation of 7.07%. Combined Tangency was much less stable: its cryptocurrency weight varied from zero to 48.13%. This shows that the name "combined fund" by itself does not explain how much cryptocurrency risk an investor actually receives.

TailGuard allocated an average of only 0.59% to cryptocurrency and was never close to its 30% cryptocurrency-sleeve limit. The equity-dominated allocation was therefore created by the historical CVaR objective rather than by the cap. The practical issue I see in Figure A3 is that Combined Tangency's large allocation changes may also create higher turnover and implementation costs. Because those costs are excluded from the backtest, its reported return may be harder to achieve in practice.

2.4 Risk–Return Positioning and Fund Fact Sheets

Figure A4 plots the 14 core funds by annualised return and volatility across the three

asset universes and four portfolio methods.

The cryptocurrency funds occupy the upper-right area of Figure A4, showing that their higher returns were accompanied by much greater volatility. Crypto Minimum Volatility produced the highest return, but this must be considered with the severe drawdown shown in Figure A2.

Within the combined universe, Combined Tangency produced higher return but also higher volatility than Combined Risk Parity. Equity Equal Weight also appears above the optimised equity funds at a similar level of risk. TailGuard remains in the low-return, low-volatility area and does not improve on Combined Minimum Volatility, while Sentiment Tilt Equity appears slightly below and to the right of Equity Tangency. These positions support my view that adding a more complex objective does not automatically move a portfolio in a better direction.

Figure A4 does not capture drawdown, tail loss, concentration or implementation costs. The individual fund fact sheets provide these results with holdings and weight history, so a fund should not be judged from the risk–return map alone.

3. Finance-Aware Sentiment Index

3.1 Headline Preparation and VADER Scoring

The sentiment calculations are based on the official headlines data of the 50 equities. Cryptocurrency is omitted since there is no cryptocurrency news in the headlines dataset. After excluding 2,847 exact headline duplicates based on ticker symbol, date and title, there remain 146,836 unique headlines. Headlines published on the same date for the same ticker may be different because they can be about different information events.

The headline dates are stored in UTC, while dates of the equity prices are timezone-naïve. All dates are normalised prior to alignment. Headlines published on an equity trading day stay there, but weekend and equity-market holiday headlines are pushed ahead to the next equity trading day. This avoids associating news with a time when the equity market was not open.

Each headline is evaluated with Plain VADER and Finance-Extended VADER separately prior to aggregation and combination. Capitalisation, punctuation, intensifiers, negation and stopwords are preserved since they are used by VADER when calculating sentiment (Hutto & Gilbert 2014). Compound scores above or equal to 0.05 are positive, scores below or equal to -0.05 are negative and the remainder are neutral. Headline sentiment scores are then averaged into 37,962 ticker-day observations before being combined into sector indices. Averaging sentiment scores

of headlines separately prevents artefacts from joining several headlines into an artificially long text when doing sentiment calculations. For portfolio fusion, the finished sector signal is lagged by one equity trading day, so that a weekend headline mapped forward to Monday can first begin to affect the Tuesday decision.

3.2 Finance Lexicon Extension and Diagnostic Evaluation

Plain VADER may not be adequate for finance texts, since general-purpose sentiment dictionaries are likely to mislabel common words used in finance texts (Loughran and McDonald, 2011). A transparent finance lexicon is therefore added to the model, extending it with a 28-term dictionary. Examples of positive terms are “outperform”, “upgrade” and “surge”, while negative terms are “underperform”, “downgrade”, “default” and “bankruptcy”. Only the selected word valences are changed, while the default VADER rules on punctuation, capitalisation, intensifiers and negation remain unchanged for the extension. I kept the extension as compact as possible so that every added term and its effect could be inspected and not just used as a black box for the model (Consoli, Barbaglia and Manzan, 2022).

As listed in Table A2, the finance extension altered 5,756 of the 37,962 ticker-day compound scores, which is 15.16% of the sample. The mean compound sentiment rose from 0.113 to 0.122, while the share of neutral decreased from 32.26% to 30.51% and the share of positive rose from 56.30% to 58.41%. This shows that the finance lexicon did indeed affect the model, but the fact that there was a change in score distribution does not prove that the new classifications are more accurate.

Since there are no ground-truth sentiment labels in the official headlines dataset, I manually classified a reproducible sample of 150 headlines before making the VADER predictions visible. The sample includes all ten sectors, spans all four years and is composed of 68 positive, 59 neutral and 23 negative headlines. Keeping the predictions hidden during labelling reduced my risk of being influenced by the model in my decisions.

As listed in Table A3, Finance-Extended VADER raised accuracy from 46.0% to 48.0% and macro-F1 from 0.412 to 0.428. It changed only four of the 150 predicted labels: three of the changes corrected errors in the Plain VADER predictions, while none of the changes transformed a correct label into an incorrect one. The outcome is positive, but the improvement is modest.

The main weakness is in the recognition of negative headlines. Both models correctly identified only 21.7% of the 23 manually labelled negative headlines.

Finance-Extended VADER produced slightly higher neutral and positive recall, but the sample is limited to 150 labels assigned by one evaluator. My conclusion is thus not

robust to any great degree: the finance lexicon improved classification modestly, but the evidence does not seem strong enough to label the model as reliable, at least in relation to negative financial language.

3.3 Sector Sentiment Index and News Coverage

The ticker-day observations are aggregated into a daily sentiment index of each of the ten equity sectors. Each ticker is given equal weight in its sector, so that firms with more headlines do not dominate the result. Ticker-days with no headlines receive a neutral compound value of zero. This delivers a complete panel of daily data, although it also draws the less news-covered sectors towards the neutral level.

For presentation purposes, the compound is rescaled from -1 to $+1$ onto a 0–100 scale, where 50 represents a neutral sentiment. Figure A5 reports a 21-trading-day moving average in order to reduce daily noise and make the sector series easier to compare. The smoothed measure is employed for presentation purposes only: the portfolio-fusion test in Section 4.1 is based on the lagged and causally standardised signal. Aggregating the scores of the text into a news-sentiment time series is in line with a large body of research in the news-sentiment literature (Shapiro, Sudhof and Wilson, 2022), although the particular equal-weighted construction of sectoral indices and the 0–100 presentation used are specific to MarketPulse.

The sector indices stayed above 50, on average. Utilities had the highest full-sample mean at 55.61 and Technology at 55.51, while Materials recorded the lowest of 52.90. The relatively narrow differences suggest a generally positive headline base instead of large and stable sentiment differences between sectors.

News coverage varied far more than the mean sentiment scores. Consumer had a mean ticker coverage of 93.92%, followed by Financials at 87.50% and Technology at 86.46%. Materials and Real Estate had 53.12% and 55.59%. Since a no-news observation is assigned a neutral score, the sectors with fewer headlines are pulled more towards 50. The point I draw from this is the risk that a score close to 50 may be the result of little news rather than an expression of strong agreement that the news that was available was neutral.

The sector index should thus be interpreted in relation to the coverage data. It is a descriptive measurement of headline tone and not a forecast of returns. Section 4 separately tests whether the lagged signal added value when applied to equity-fund weights.

4. Sentiment Fusion and Innovation

This chapter considers two such extensions: Sentiment Tilt Equity, which combines

equity returns with lagged sector-news sentiment, and MarketPulse TailGuard, which incorporates both historical CVaR and an aggregate cryptocurrency cap. Again, neither is assumed to outperform. They are simply compared with simpler benchmarks.

4.1 Look-Ahead-Safe Sentiment-Fusion Design

The core fusion study combines Finance-Extended sector sentiment with Equity Tangency weights. Equity Tangency and Sentiment Tilt Equity share the same equity universe, 252-observation window, monthly rebalance dates, zero risk-free rate and long-only constraints. The only difference is the sentiment adjustment. As such, the before-and-after comparison isolates the effect of that adjustment alone. The same fixed rule is then reapplied to Equal Weight, Minimum Volatility and Risk Parity as robustness checks.

To avoid look-ahead bias, sector sentiment is lagged by one equity trading day. A weekend headline mapped to Monday can therefore first affect Tuesday's signal. The lagged score is standardised using an expanding sector-specific mean and standard deviation calculated only from information available at that time. At least 20 observations are required, and the causal z-score is clipped between -3 and $+3$:

$$\begin{aligned} z_{s,t}^{causal} \\ = \text{clip} \left(\frac{S_{s,t-1} - \bar{S}_{s,t-1}^{exp}}{\sigma_{s,t-1}^{exp}}, \right. \\ \left. -3, 3 \right) \end{aligned} \quad (7)$$

At each monthly rebalance, the causal scores are standardised again across the ten sectors so that the tilt reflects relative sentiment at that date. Each sector receives a multiplier equal to one plus 0.30 times its cross-sectional score, restricted to the range 0.50–1.50:

$$m_{s,t} = \text{clip} \left(1 + 0.30 \tilde{z}_{s,t}, 0.50, 1.50 \right) \quad (8)$$

Each equity's base weight is multiplied by its sector multiplier and then renormalised:

$$\tilde{w}_{i,t} = \frac{w_{i,t}^{base} m_{s(i),t}}{\sum_j w_{j,t}^{base} m_{s(j),t}} \quad (9)$$

In Equations (7)–(9), S is the lagged sector sentiment score, z is the causally standardised score, m is the sector multiplier, and w represents the base and tilted

portfolio weights. After renormalisation, the 20% individual-asset cap is reapplied and any excess weight is redistributed across assets with remaining capacity.

The sentiment rule tilts the portfolio, in the above fashion, to higher weights in sectors with relatively positive lagged sector-news tone and lower weights in sectors with relatively negative sector-news tone. The sentiment rule tests the hypothesis that there is predictive information in relative sector-news tone about subsequently realised returns. There is prior evidence linking media sentiment about business news with stock-market outcomes (Tetlock 2007), but there is no need in this analysis to suppose that this relationship will exist in this sample; Sections 4.2 and 4.3 examine the relationship against matched base funds.

4.2 Core Before-and-After Fusion Result

Table A4 compares Equity Tangency with its sentiment-tilted counterpart. The estimation window, asset universe, rebalance dates and constraints are unchanged, so the performance difference isolates the sentiment rule.

Because of the tilt, the annualised return fell from 7.26% to 6.10% and the Sharpe ratio fell from 0.486 to 0.423. Total return fell from 23.29% to 19.37% and the maximum drawdown went from -23.20% to -24.80%. CVaR loss also rose slightly from 2.51% to 2.52%. The core alteration did not boost either return or downside protection.

Figure A6 shows that the two growth paths remained relatively close, but the tilted fund ended below the base fund and fell further behind later in the sample. This implementation therefore did not add value during 2021–2023.

4.3 Sentiment-Fusion Robustness Across Four Portfolio Methods

To check whether the core result was unique to tangency optimisation, I repeated the sentiment fusion using the same signal lag, tilt strength, standardisation procedure and constraints for Equity Equal Weight, Equity Minimum Volatility, Equity Tangency and Equity Risk Parity.

As shown in Table A5, the tilt reduced annualised return by 0.92–1.73 percentage points and the Sharpe ratio by 0.063–0.093 across all four methods. Only Minimum Volatility had a shallower maximum drawdown, improving by 1.07 percentage points, but its return and Sharpe ratio still fell. The signal therefore produced no consistent return or downside benefit.

Figure A7 shows that all four tilted funds ended below their matched base funds, especially Equal Weight and Risk Parity later in the sample. Because the result was

negative under both simple and optimised methods, I relate the weakness more to the fixed-strength sentiment rule than to one optimiser.

These findings do not show that news sentiment is irrelevant to asset pricing. They show only that this sector-level, fixed-strength and monthly rebalanced implementation did not add value during 2021–2023. Firm-level or faster signals could be tested in future using pre-specified rules.

4.4 MarketPulse TailGuard: Conditional Value at Risk Innovation

Variance treats positive and negative deviations symmetrically, although investors are generally more concerned about large losses. MarketPulse TailGuard therefore minimises historical 95% Conditional Value at Risk (CVaR), defined as the average loss among the worst 5% of return observations (Rockafellar and Uryasev, 2000). It uses the combined 60-asset universe, a 252-observation window and monthly rebalancing.

$$\min_{\mathbf{w}, \alpha} \left[\alpha + \frac{1}{(1-q)T} \sum_{\tau=1}^T \max(-\mathbf{w}^T \mathbf{r}_{\tau} - \alpha, 0) \right], \quad q = 0.95 \quad (10)$$

In Equation (10), $\mathbf{w}$ is the portfolio-weight vector, $\mathbf{r}$ is the historical asset-return vector, T is the estimation-window length, q is the 95% confidence level, and α is the estimated loss threshold. TailGuard is subject to the following constraints:

$$\mathbf{1}^T \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.20, \quad \sum_{i \in C} w_i \leq 0.30 \quad (11)$$

The constraints require a fully invested, long-only portfolio, a 20% cap on each asset and a 30% cap on total cryptocurrency exposure. The cryptocurrency cap is specific to TailGuard and is not imposed on the baseline combined funds.

Table A6 compares TailGuard with Combined Minimum Volatility and Combined Risk Parity.

Finally, TailGuard had a lower return and Sharpe ratio than Combined Minimum Volatility. It had a 3.28% annualised return versus 5.86%, and a Sharpe ratio of 0.304 versus 0.515. Its CVaR loss was greater than Combined Minimum Volatility at 1.89% versus 1.76%, and its maximum drawdown was deeper at –18.37% versus –15.77%. Combined Risk Parity had the highest return and Sharpe ratio of the three funds, though its CVaR loss and maximum drawdown were worse than TailGuard’s.

I was anticipating that direct CVaR minimisation would result in the best realised downside protection, but this is not what happened. There are only about 13

observations in the worst 5% of a 252-day window, and it is by no means clear that these historical losses will resemble the losses that occur after rebalancing. At least in this case, this helps explain why a specialised downside objective can still perform worse than the simpler minimum-volatility approach.

The average allocation to cryptocurrencies for TailGuard was only 0.59%, and its maximum allocation was 5.86%, as shown in Figure A8. Thus, the 30% cap never became binding. The CVaR objective itself, therefore, rather than the cap, produced the equity-dominated portfolio. TailGuard remains a distinct implemented innovation, but the results do not support presenting it as the best downside-protection fund in this sample.

5. Streamlit App and Investor Journey

MarketPulse Funds is a Streamlit research app for self-directed retail investors and research users. Its six tabs - Compare Funds, Fund Fact Sheets, Build an Allocation, News Pulse, Downside Innovation and Method Notes - move from comparing the 14 funds to examining individual funds and testing allocations. The app provides historical evidence rather than personal financial advice.

Compare Funds summarises return, volatility, Sharpe ratio, Sortino ratio, CVaR, maximum drawdown and total return with interactive growth and drawdown charts. Fund Fact Sheets adds holdings and weight history, while Build an Allocation normalises selected weights to 100% and compares gross and fee-adjusted historical values. The allocation tool uses only dates common to the selected funds. Crypto-only allocations retain the seven-day calendar and 365-period annualisation, while allocations containing an equity or combined fund use shared equity dates and 252-period annualisation, in accordance with the Project Brief.

News Pulse reports sector sentiment, news coverage, lexicon effects and the core fusion result, while Downside Innovation reports TailGuard and the four-method robustness test. Method Notes summarises the main assumptions and limitations. For reliable deployment, the app uses precomputed CSV artifacts rather than rerunning VADER or the backtests. When testing it on my side, I found that fixed CSS colours reduced Dark Mode readability, so I switched them to Streamlit theme variables for both Light and Dark modes.

6. Critical Reflection and Three Recommendations

6.1 Critical Reflection

The findings indicate that greater complexity did not necessarily improve out-of-sample performance. Equity Equal Weight produced a higher return and Sharpe ratio than the optimised equity methods. In my judgment, its simplicity may have helped because it does not rely on expected-return and covariance estimates, which can be unstable in a relatively short rolling window (DeMiguel, Garlappi & Uppal 2009).

The fund results also show why I would not select a portfolio using one performance measure alone. Combined Tangency had the highest return among the equity and combined funds, Combined Risk Parity the highest Sharpe ratio, and the minimum-volatility funds shallower drawdowns. Crypto Minimum Volatility is the clearest example: despite its strong return and Sharpe ratio, its -74.47% maximum drawdown still involved substantial loss. The combined-fund results should also be interpreted under the Brief's left-merge convention, which excludes weekend-only cryptocurrency returns.

I had anticipated that TailGuard's CVaR objective would provide stronger realised downside protection, yet it performed worse than Combined Minimum Volatility on CVaR and maximum drawdown. This shows that historical downside-risk optimisation does not guarantee better subsequent protection. Similarly, better headline classification did not produce a better trading signal, and the fixed sentiment tilt reduced return and Sharpe ratio across all four methods.

These findings are limited to 2021–2023 and exclude transaction costs, taxes, slippage and market impact. The news data contains no cryptocurrency headlines, no-news observations are treated as neutral, and manual validation uses 150 labels from one evaluator. I therefore view the results as historical evidence, not forecasts.

6.2 Recommendation 1: Add Turnover and Implementation Costs

An upcoming revision ought to compute turnover at each monthly rebalance and transform the current gross backtest into a gross-versus-net set of results. Equity and cryptocurrency turnover should be assessed independently due to potential variations in transaction costs, liquidity and slippage between the two. Every fund fact sheet may then summarise average monthly turnover, estimated transaction-cost drag, gross and net returns, and gross and net Sharpe ratios.

This approach would be especially important for Combined Tangency, whose cryptocurrency exposure fluctuated between 0% and 48.13% across monthly rebalances. As asset-class weights shift dramatically, substantial trading could result even if individual weights do not exceed the 20% allocation cap. I would extend the allocation tool's existing management-fee comparison by introducing separate equity

and cryptocurrency trading-cost assumptions under low, central and high scenarios. A consequent evaluation would demonstrate the extent to which fund rankings remain credible after transaction costs instead of assuming that investors are able to fully reproduce the reported gross returns.

6.3 Recommendation 2: Expand Sentiment Validation and Improve Negative Recall

The manual validation exercise spanning 150 headlines should be extended into a larger and more evenly distributed sample covering every sector and year. Each headline ought to be evaluated by at least two people who independently label it, with disagreements reviewed before creating the final reference label. The evaluation exercise should report inter-rater agreement, macro-F1 and class-specific precision and recall in addition to overall accuracy.

My initial modelling focus would centre on negative-headline recall given that Finance-Extended VADER identified only 21.7% of the 23 manually labelled negative headlines. I would revisit the unrecognised headlines to ascertain whether the errors stemmed from finance-specific vocabulary, negation, contextual usage or headlines that reference multiple events. Previously labelled data might be utilised for lexicon development, a separate period for validation and the latest period as an untouched holdout. The lexicon, tilt strength and signal frequency would then be fixed before comparing daily, weekly and monthly portfolio rebalancing. Such separation would minimise the risk of selecting model settings only because they performed well in the final investment sample.

6.4 Recommendation 3: Add Look-Through Risk Controls and Stress Testing

The allocation tool should compute the investor's underlying exposures rather than showing selected fund weights only. I am worried that several funds can have the same holdings and a multi-fund allocation can then seem diversified yet concentrated in specific equities, cryptocurrencies or sectors. The app should aggregate the latest holdings and report look-through asset-class, sector and individual-asset weights.

The user could then specify research limits on cryptocurrency exposure, annualised volatility, CVaR and maximum drawdown. The app should warn if any limit is breached while making it clear that the warning is based on historical evidence and not personal financial advice. Historical stress tests could also report the dollar loss, percentage loss and time to recovery under the worst observed equity and cryptocurrency periods, including the 2022 cryptocurrency decline.

TailGuard shows why one portfolio constraint is insufficient to show protection. Its 30% cryptocurrency limit was never binding, yet it still performed worse than

Combined Minimum Volatility on realised downside measures. I would therefore show look-through exposure, stress losses and several risk measures together rather than relying on a fund name or one optimisation objective to communicate risk.

Conclusion

MarketPulse Funds created 14 systematic funds across equity, cryptocurrency and combined universes using monthly walk-forward evaluation. The findings show that return, risk-adjusted performance and downside protection are different objectives. Cryptocurrency funds delivered the highest returns but also large drawdowns, while the equity and combined funds performed differently depending on the metric.

Neither extension clearly improved performance. Finance-Extended VADER slightly improved manual-validation results, but the sentiment tilt reduced return and Sharpe ratio across all four methods, while TailGuard failed to outperform Combined Minimum Volatility on realised downside protection. I therefore view these negative findings as evidence that greater model complexity should be justified by out-of-sample results rather than assumed to be better. MarketPulse remains a reproducible historical research tool rather than a forecast or source of personal financial advice.

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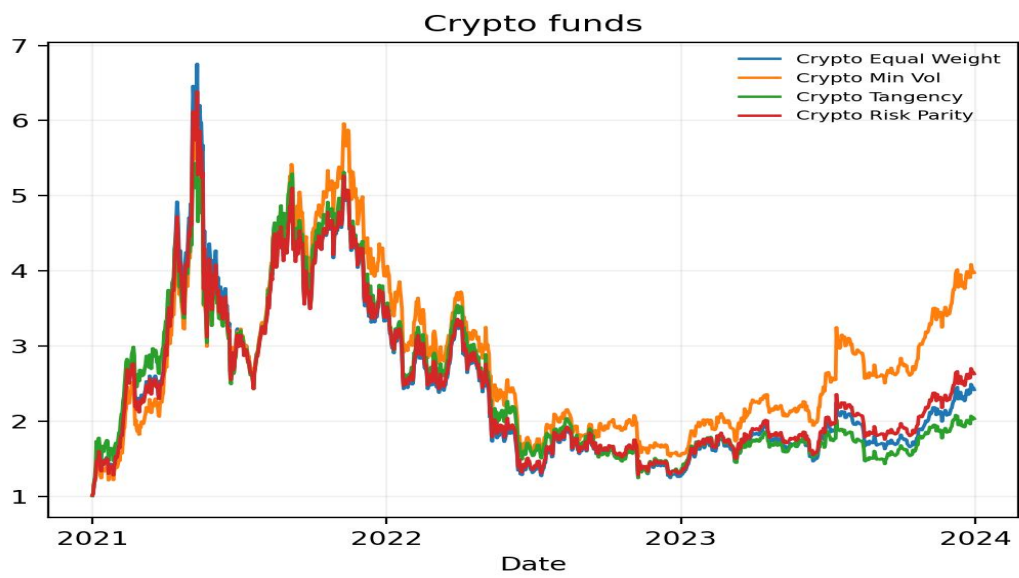
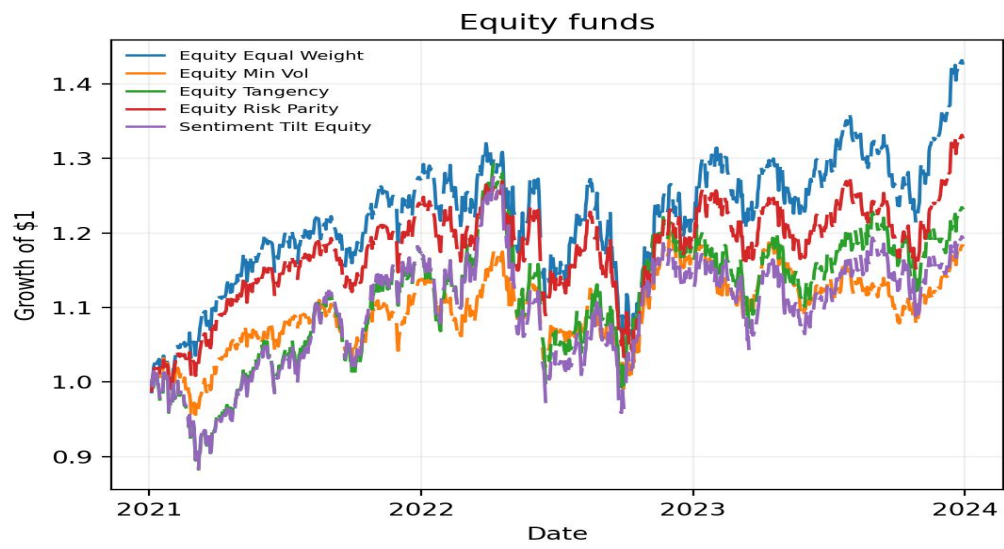
Appendix A — Exhibits

Table A1. Out-of-sample fund performance, 2021–2023

Fund	Annualised Return	Annualised Volatility	Sharpe Ratio	Maximum Drawdown
Equity Equal Weight	12.64%	16.17%	0.817	–20.32%
Equity Minimum Volatility	5.79%	12.60%	0.510	–15.62%
Equity Tangency	7.26%	17.61%	0.486	–23.20%
Equity Risk Parity	9.97%	14.61%	0.724	–18.55%
Crypto Equal Weight	34.19%	80.42%	0.772	–81.60%
Crypto Minimum Volatility	58.40%	76.63%	0.986	–74.47%
Crypto Tangency	26.65%	76.71%	0.694	–77.12%
Crypto Risk Parity	37.99%	78.47%	0.807	–79.94%
Combined Equal Weight	14.98%	21.25%	0.763	–28.75%
Combined Minimum Volatility	5.86%	12.61%	0.515	–15.77%
Combined Tangency	19.06%	23.41%	0.862	–23.14%
Combined Risk Parity	13.97%	16.10%	0.893	–19.98%
Sentiment Tilt Equity	6.10%	17.69%	0.423	–24.80%
MarketPulse TailGuard	3.28%	13.72%	0.304	–18.37%

Note: Annualised return is the compound annual growth rate. Sharpe ratios assume a zero annual risk-free rate. Equity-only and combined funds use 252 periods per year, while crypto-only funds use 365. Maximum drawdown is the largest peak-to-trough decline in the growth-of-one-dollar series.

Source: Author's calculations using the official FINS3645 project data.



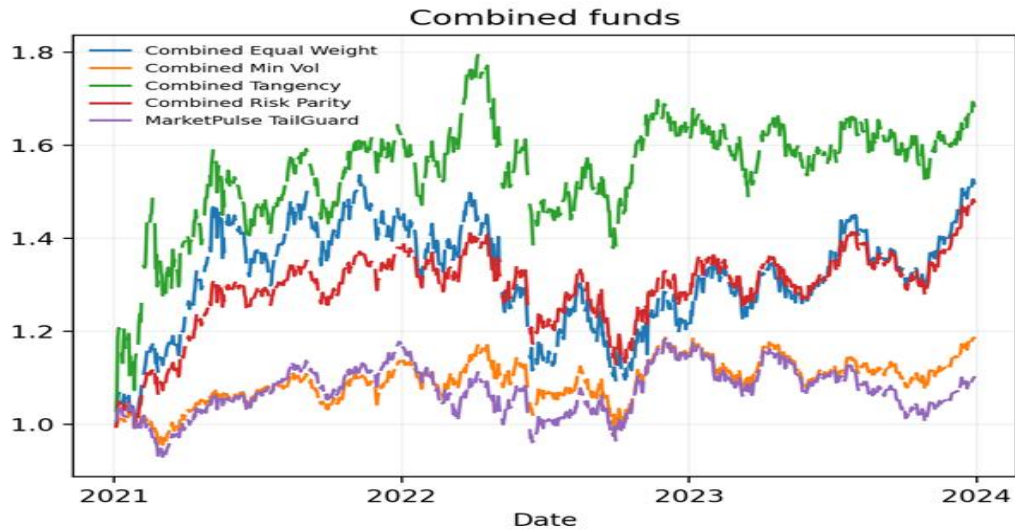
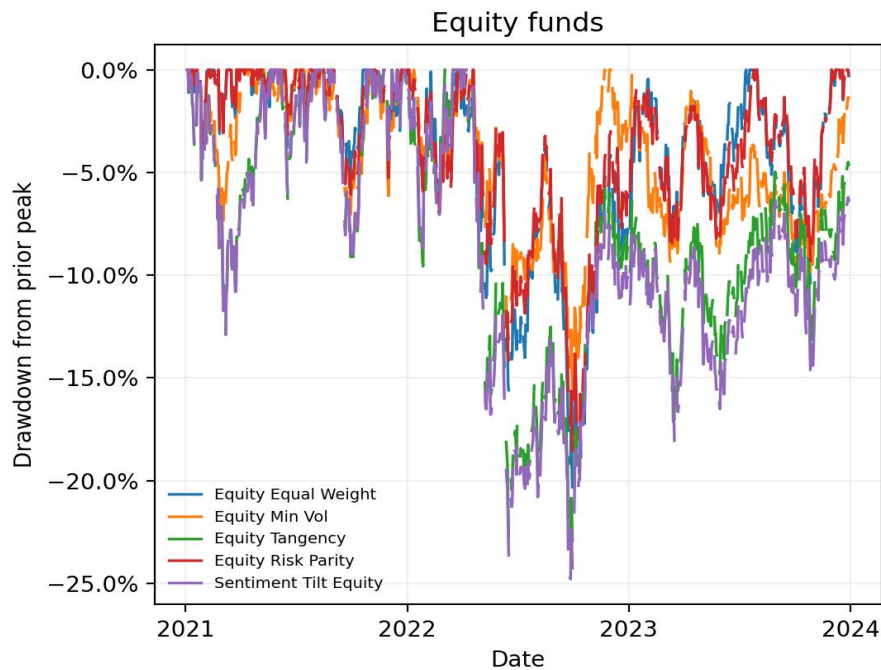


Figure A1. Out-of-sample growth of \$1 for the 14 core funds by investment universe, January 2021–December 2023.

Note: Equity-only and combined funds use the equity trading calendar, while crypto-only funds retain the seven-day cryptocurrency calendar. The three panels use different vertical scales. Returns exclude transaction costs, taxes, slippage and market impact.

Source: Author's calculations using the official FINS3645 project data.



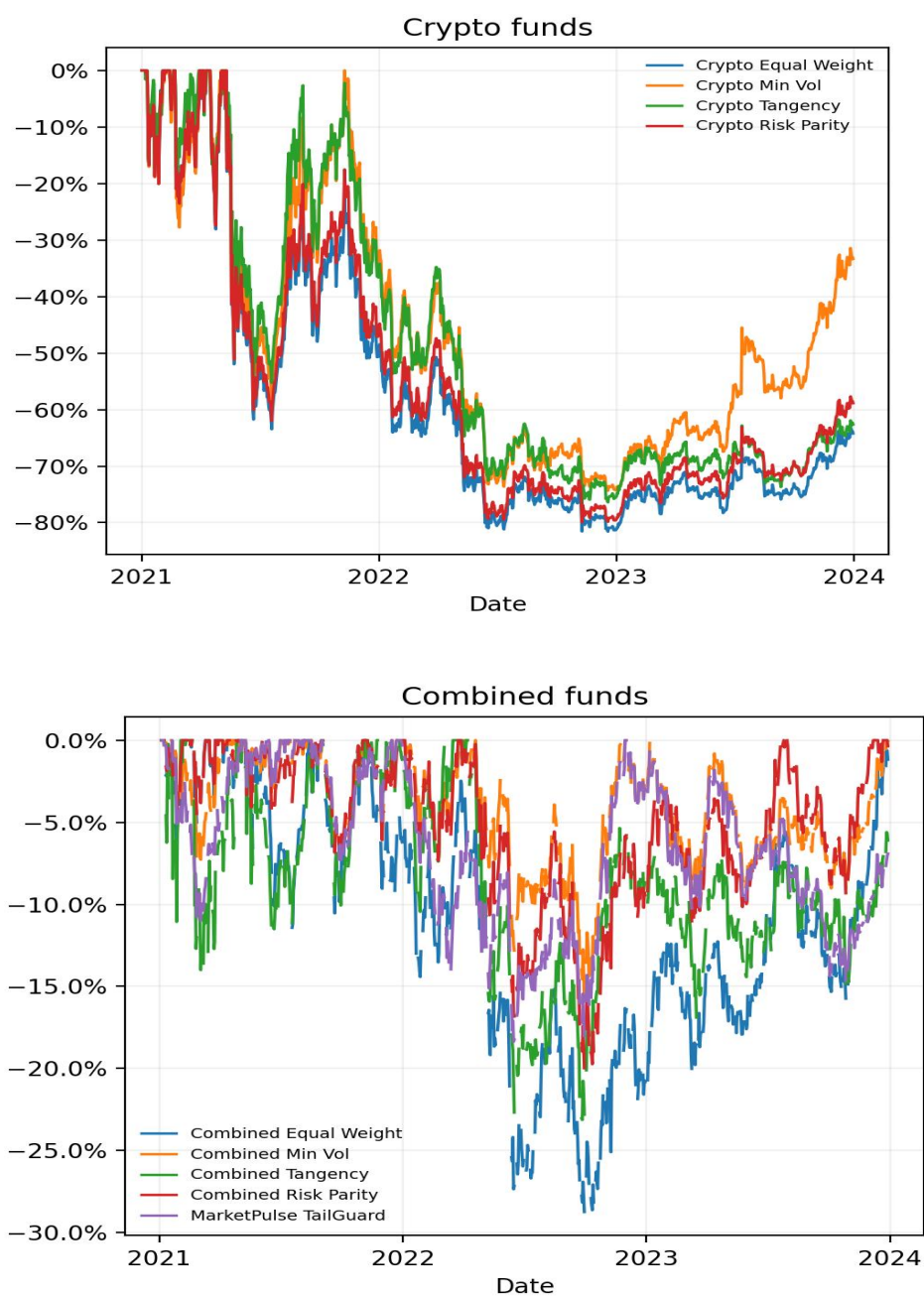


Figure A2. Out-of-sample drawdowns for the 14 core funds by investment universe, January 2021–December 2023.

Note: Drawdown is measured as the percentage decline from each fund's previous running peak. A value of zero indicates a new peak. The three panels use different vertical scales.

Source: Author's calculations using the official FINS3645 project data.

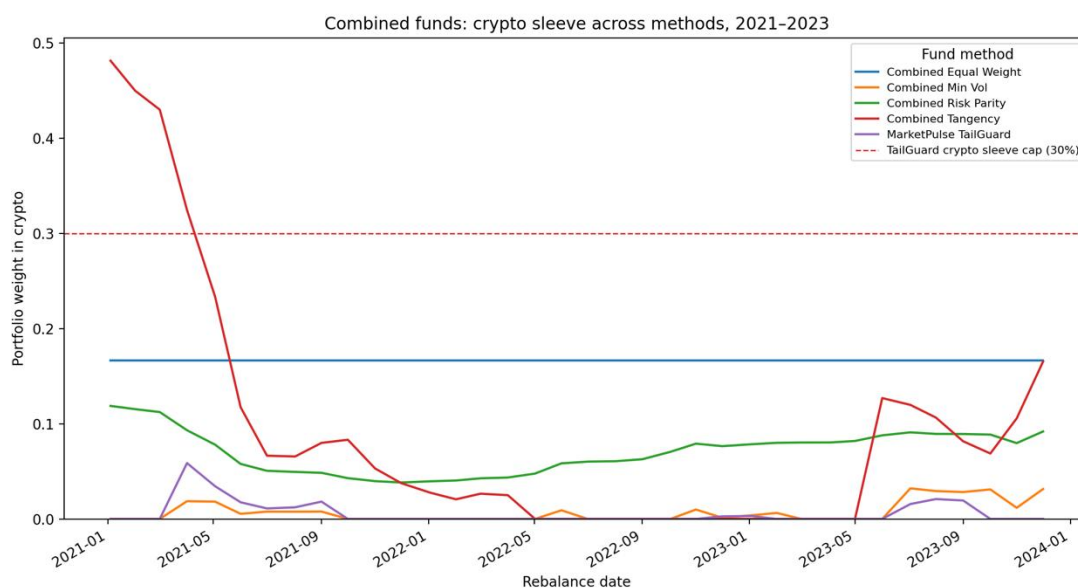


Figure A3. Total cryptocurrency allocation of the combined funds across portfolio methods, January 2021–December 2023. Each line reports the total weight allocated to the ten cryptocurrencies at a monthly rebalance. The remaining weight is invested in equities.

Note: The 20% maximum-weight constraint applies to each individual asset. The dashed 30% aggregate cryptocurrency-sleeve cap applies only to MarketPulse TailGuard; it does not apply to the four baseline combined funds.

Source: Author's calculations using the official FINS3645 project data.

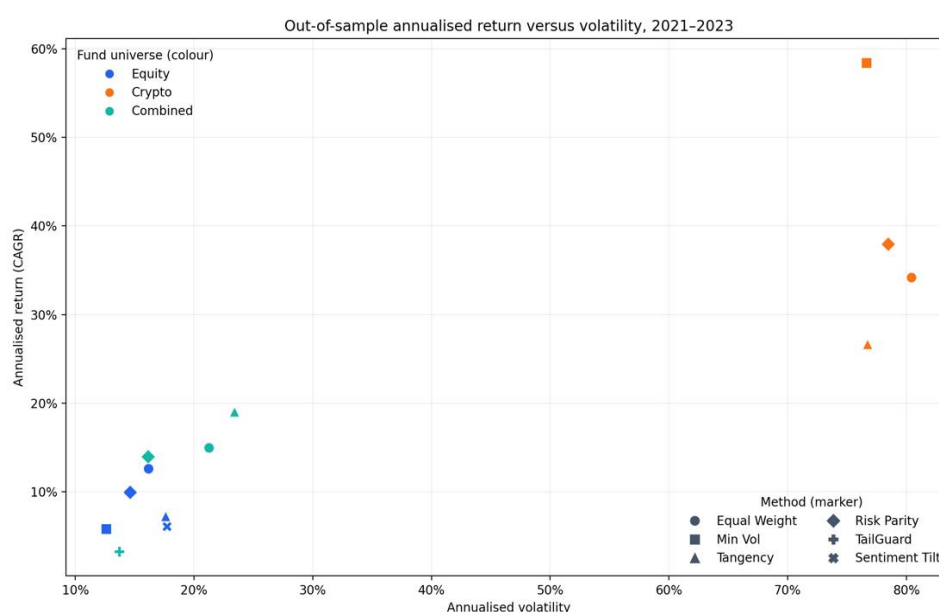


Figure A4. Out-of-sample annualised return and volatility of the 14 core funds, January 2021–December 2023. Colour represents the fund universe, while marker shape represents

the portfolio method.

Note: Annualised return is the compound annual growth rate. Equity-only and combined funds use 252 periods per year, while crypto-only funds use 365. The figure reports gross historical model returns before transaction costs, taxes, slippage and market impact.

Source: Author's calculations using the official FINS3645 project data.

Table A2. Full-sample comparison of Plain and Finance-Extended VADER on equity ticker-days, 2020–2023

Model	Mean Compound	Negative	Neutral	Positive	Compound Score Changed
Plain VADER	0.113	11.44%	32.26%	56.30%	N/A
Finance-Extended VADER	0.122	11.07%	30.51%	58.41%	15.16%

Note: Both models are evaluated on the same 37,962 ticker-day observations. Positive observations have compound scores of at least 0.05, negative observations have scores of –0.05 or below, and the remaining observations are neutral. “Compound Score Changed” reports the share of ticker-days for which the finance lexicon changed the Plain VADER compound score.

Source: Author's calculations using the official FINS3645 project data.

Table A3. Blind manual validation of headline-sentiment classifications (n = 150)

Model	Accuracy	Macro-F1	Negative Recall	Neutral Recall	Positive Recall
Plain VADER	46.0%	0.412	21.7%	54.2%	47.1%
Finance-Extended VADER	48.0%	0.428	21.7%	55.9%	50.0%

Note: The manually labelled sample contains 23 negative, 59 neutral and 68 positive headlines. Accuracy is the share of all headlines classified correctly. Macro-F1 gives equal importance to the F1 score of each sentiment class, while recall measures the share of the manually labelled observations in each class identified correctly by the model.

Source: Author's manual classification and calculations using the official FINS3645 project data.

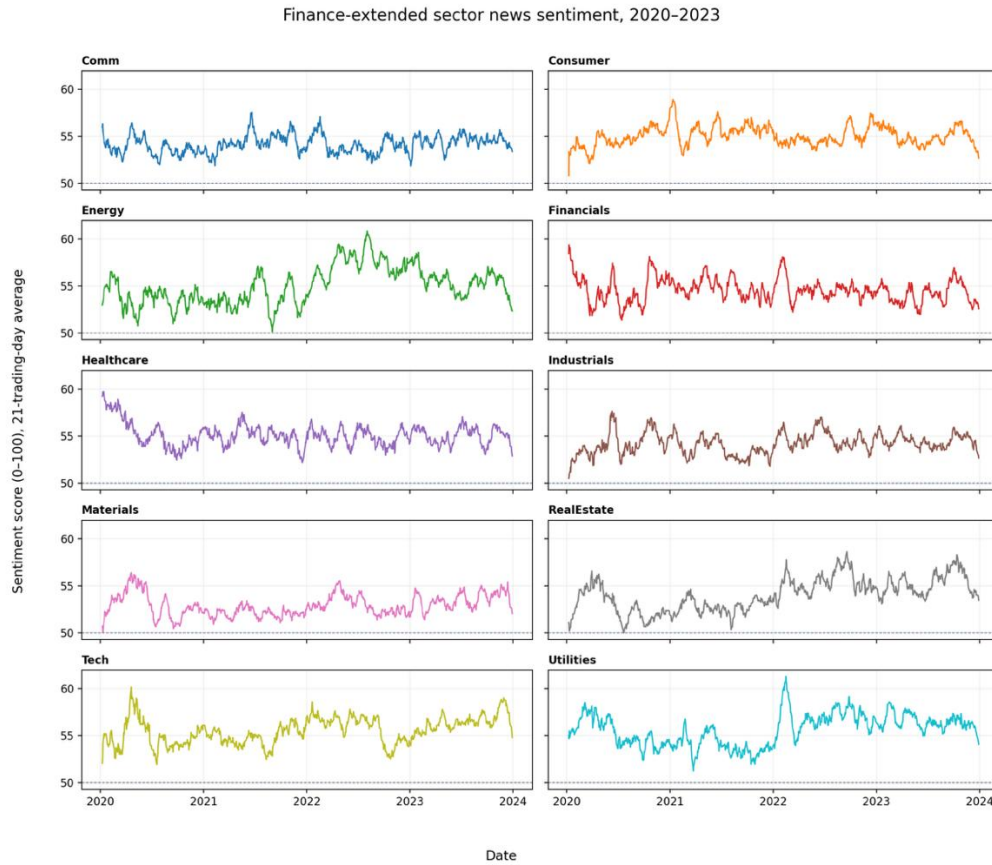


Figure A5. Finance-Extended sector news-sentiment indices, January 2020–December 2023. Each panel reports the 21-trading-day moving average of the sector sentiment score on a 0–100 scale. The dashed line at 50 represents neutral sentiment.

Note: Headline scores are first averaged to ticker-day observations and then equal-weighted across tickers within each sector. Ticker-days without headlines are assigned a neutral compound score of zero before the 0–100 transformation.

Source: Author's calculations using the official FINS3645 project data.

Table A4. Before-and-after performance of the core equity sentiment-fusion extension, January 2021–December 2023.

Fund	Annual return	Annual volatility	Sharpe ratio	Sortino	95% CVaR	Max drawdown	Total return
Equity Tangency	7.26%	17.61%	0.486	0.679	2.51%	–23.20%	23.29%
Sentiment Tilt Equity	6.10%	17.69%	0.423	0.591	2.52%	–24.80%	19.37%
Tilt Minus Base	–1.16 pp	+0.08 pp	–0.063	–0.088	+0.01 pp	–1.60 pp	–3.93 pp

Note: Both funds contain 753 observations and use 252 periods per year. “Tilt Minus Base” is

calculated as Sentiment Tilt Equity minus Equity Tangency. “pp” means percentage points. CVaR is reported as the positive magnitude of the average daily loss within the worst 5% of return observations. A positive CVaR difference and a negative maximum-drawdown difference both represent a deterioration.

Source: Author’s calculations using the official FINS3645 project data.

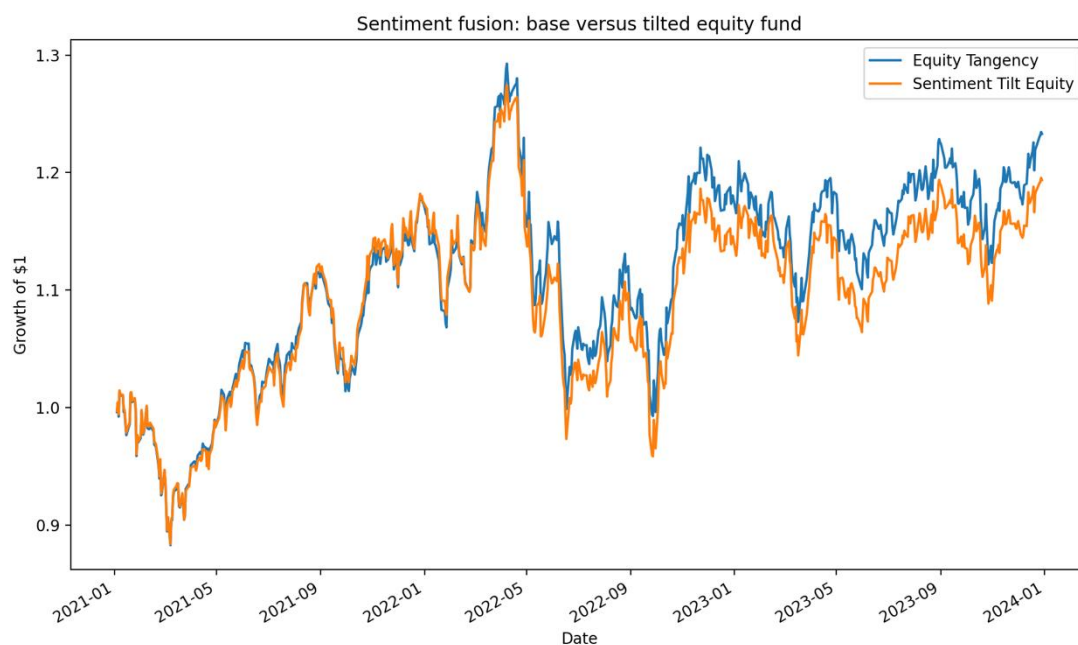


Figure A6. Growth of \$1 for Equity Tangency and Sentiment Tilt Equity, January 2021–December 2023.

Note: Both funds are rebalanced monthly using a 252-observation trailing window. Sentiment Tilt Equity applies the one-trading-day-lagged and causally standardised sector-sentiment signal to the Equity Tangency weights. Returns exclude transaction costs, taxes, slippage and market impact.

Source: Author’s calculations using the official FINS3645 project data.

Table A5. Robustness of the fixed sentiment tilt across four equity portfolio methods, January 2021–December 2023.

Base Method	Annual return	Tilted Annual Return	Return Difference	Sharpe Difference	Max-Drawdown Difference
Equal Weight	12.64%	10.92%	−1.73 pp	−0.093	−0.43 pp
Minimum Volatility	5.79%	4.87%	−0.92 pp	−0.069	+1.07 pp

Tangency	7.26%	6.10%	-1.16 pp	-0.063	-1.60 pp
Risk Parity	9.97%	8.49%	-1.48 pp	-0.092	-0.16 pp

Note: Differences are calculated as the sentiment-tilted fund minus its matched base fund. “pp” means percentage points. A positive maximum-drawdown difference indicates a shallower drawdown after the tilt, while a negative difference indicates a deeper drawdown.

Source: Author’s calculations using the official FINS3645 project data.

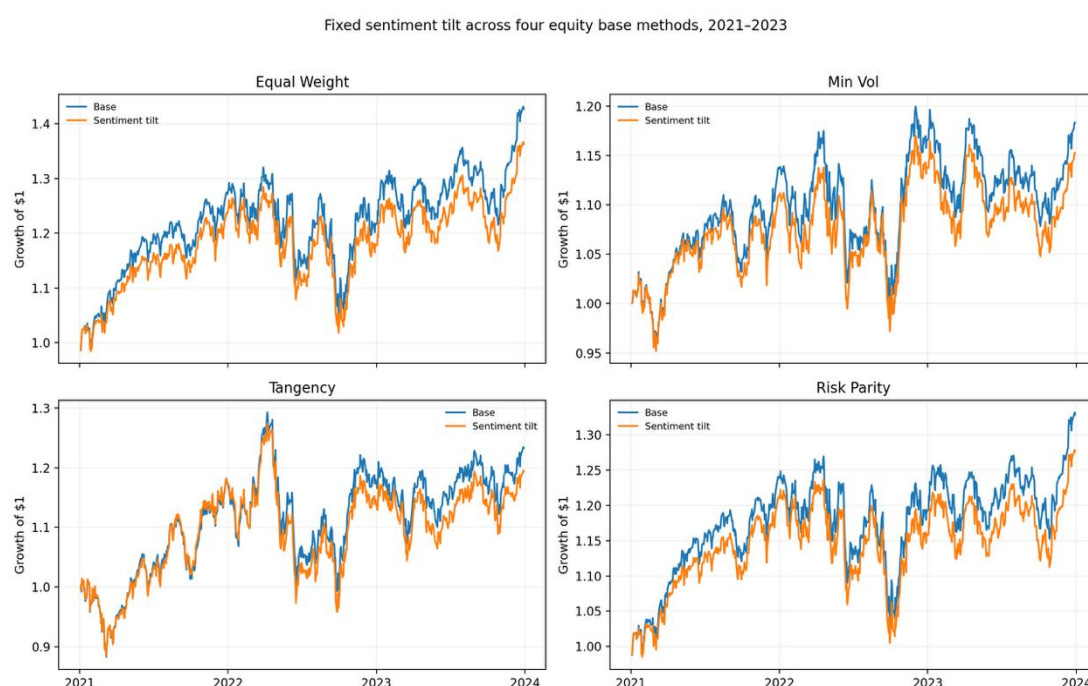


Figure A7. Growth of \$1 before and after applying the fixed sector-sentiment tilt across four equity portfolio methods, January 2021–December 2023.

Note: Each panel compares one equity base fund with a matched fund using the same one-trading-day-lagged sentiment signal, tilt strength of 0.30 and 20% individual-asset cap. The four panels use different vertical scales.

Source: Author’s calculations using the official FINS3645 project data.

Table A6. Out-of-sample comparison of MarketPulse TailGuard and combined-fund benchmarks, January 2021–December 2023

Fund	Annual	Annual	Sharpe	Sortino	95% Daily	Max	Total
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	Return	Volatility			CVaR	Drawdown	Return
Combined Minimum Volatility	5.86%	12.61%	0.515	0.731	1.76%	-15.77%	18.55%
Combined Risk Parity	13.97%	16.10%	0.893	1.288	2.28%	-19.98%	47.81%
MarketPulse TailGuard	3.28%	13.72%	0.304	0.425	1.89%	-18.37%	10.12%

Note: CVaR is the positive magnitude of the average daily loss among returns at or below the fifth percentile. It is a daily tail-loss measure and is not annualised. All three funds use 753 out-of-sample equity-calendar observations and exclude transaction costs, taxes, slippage and market impact.

Source: Author's calculations using the official FINS3645 project data.

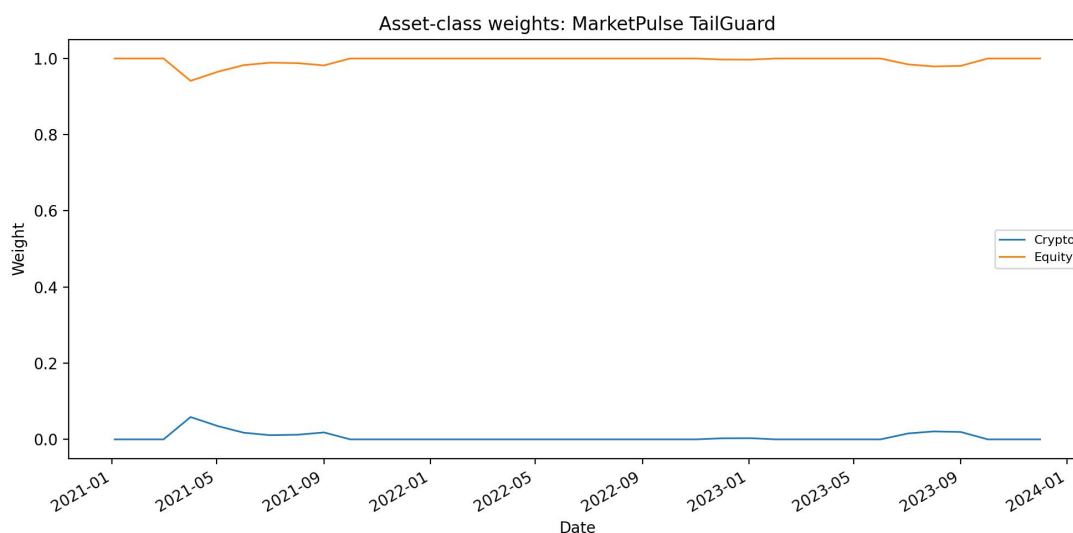


Figure A8. Equity and cryptocurrency target weights of MarketPulse TailGuard at monthly rebalances, January 2021–December 2023.

Note: The y-axis reports portfolio weights as decimals, so 0.06 represents a 6% allocation. Across monthly rebalances, TailGuard's cryptocurrency weight averaged 0.59% and reached a maximum of 5.86%. The figure reports aggregate asset-class weights rather than individual holdings. TailGuard is subject to a 20% cap on each asset and a 30% cap on the combined cryptocurrency sleeve.

Source: Author's calculations using the official FINS3645 project data.